3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

The Improved Teeny-TOF Mass Spectrometer for Chemical and Biological Sensing

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ABSTRACT

A miniature, low voltage, reflectron time-of-flight mass spectrometer has been developed and tested on a series of compounds ranging in mass from a few hundred Da to over 50kDa. The design employs a small and commercially available 10 l/s pump, a 140uJ nitrogen laser, and fiber optics to deliver the laser energy. A gridless, focusing ion source is operating at 4.5kV, the ruggedized reflector is rolled from a flexible circuit board and encased in fiberglass, and the detector has an improved anode that reduces ringing commonly found in coaxial designs. The first prototype is currently being tested for its use in chemical and biological threat detection.